

Yu-chao Guo

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Address: No. 2399 Hongguang Avenue, Chang'an District, Xi'an, Shaanxi Province, (Innovation Harbor Campus)

Education

Sep. 2026 – Present	Ph.D. Student in Chemical Engineering and Technology, Xi'an Jiaotong University, Advisor: Prof. Hang Zhang
Aug. 2023 – Jun. 2026	M.S. in Chemistry, emphasis in Polymer chemistry and physics, Jilin University, Advisor: Prof. Wenke Zhang, Thesis Title: Study on the Effect of Chain Entanglement on the Mechanochemical Behavior of Polymers.
Aug. 2019 - Jun. 2023	B.S. in Chemistry, emphasis in Organic chemistry, Northeast Agricultural University

Research experience

Sep. 2026 – Present	Ph.D. Student in Xi'an Jiaotong University, Objective: Advisor: Prof. Hang Zhang
Aug. 2023 – Jun. 2026	M.S. in Jilin University, Objective: Utilizing spiropyran mechanophores as crosslinkers to characterize stress transfer in polymer networks, and enhancing the activation efficiency of mechanophores in bulk materials from 1% to 5%. Advisor: Prof. Wenke Zhang (The National Science Fund for Distinguished Young Scholars)
Aug. 2019 - Jun. 2023	B.S. in Chemistry, Northeast Agricultural University, Objective: Synthesis and bioactivity evaluation of potential dichloroacetamide oxazolidine herbicide safeners. Advisor: Assoc. Prof. Yingjiao Li

Publications

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2. Yu-chao Guo, Ziwen Ma, Wenke Zhang[†]. “Visualizing Stress Transfer in Polymer Networks via Mechanophore Activation” manuscript in preparation.
1. Ma, Zhao-Yu, Lu, Wan, Guo, Yu-Chao and Li, Ying-Jiao[†]. "Crystal structure of 2,2-dichloro-1-(4-chloro-1H-indol-1-yl)ethan-1-one, C₁₀H₆Cl₃NO" Zeitschrift für Kristallographie - New Crystal Structures, 237, 4, 2022, 593-594. <https://doi.org/10.1515/ncrs-2022-0117>

Academic Activities

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- 2023 National Postgraduate Summer Workshop on “Supramolecular Chemistry and Self-Assembly”, *Jilin University*.
Lecturers: Academicians Xi Zhang, Mei-Xiang Wang, Li-Feng Chi, Zhong-Qun Tian, He Tian, Li-Zhu Wu; Professors Xin Xu (*Fudan University*), Xiao-Dong Chen (*Nanyang Tech University*)

Honors and Awards

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- 2024 Postgraduate Academic Scholarship, Jilin University.

Skills and Interests

Skills:

- Organic Synthesis: Air-free techniques (Schlenk line), glovebox, etc. Polymerization: Free radical polymerization, SET-LRP, etc. Polymer Characterization: Universal testing machine, rheology, etc.
- English: CET-4; preparing for IELTS.

Interests:

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1. Polymer Mechanochemistry
 2. Polymer Network Topology Engineering
 3. Mechanochemical Toughening & Resilient Elastomers